Eduardo Ruiz Robles 13/02/2026

# Questions

(passed/failed)

**1) General outcome of the simulation:**

**- How many integration steps were needed to run the 2-nanosecond simulation?**

The 2ns simulation required 1,000,000 integration steps, since a timestep of 2 fs (0.002 ps) was used.

**- How many atoms were removed from the system?**

A total of 2,214 water molecules were removed, corresponding to 6,642 atoms removed from the system. Each water molecule has 3 atoms.

**- What is the performance gain upon removal of the water (tip: check the Performance line in the md.log files)?**

The removal of water molecules resulted in a significant performance gain of 56.754 ns/day, representing a 131.7% increase in simulation speed.

```
(base) eduardoruiz@192 md_vacuum % grep "Performance" md.log
Performance:       99.833        0.240
(base) eduardoruiz@192 md_vacuum %
```

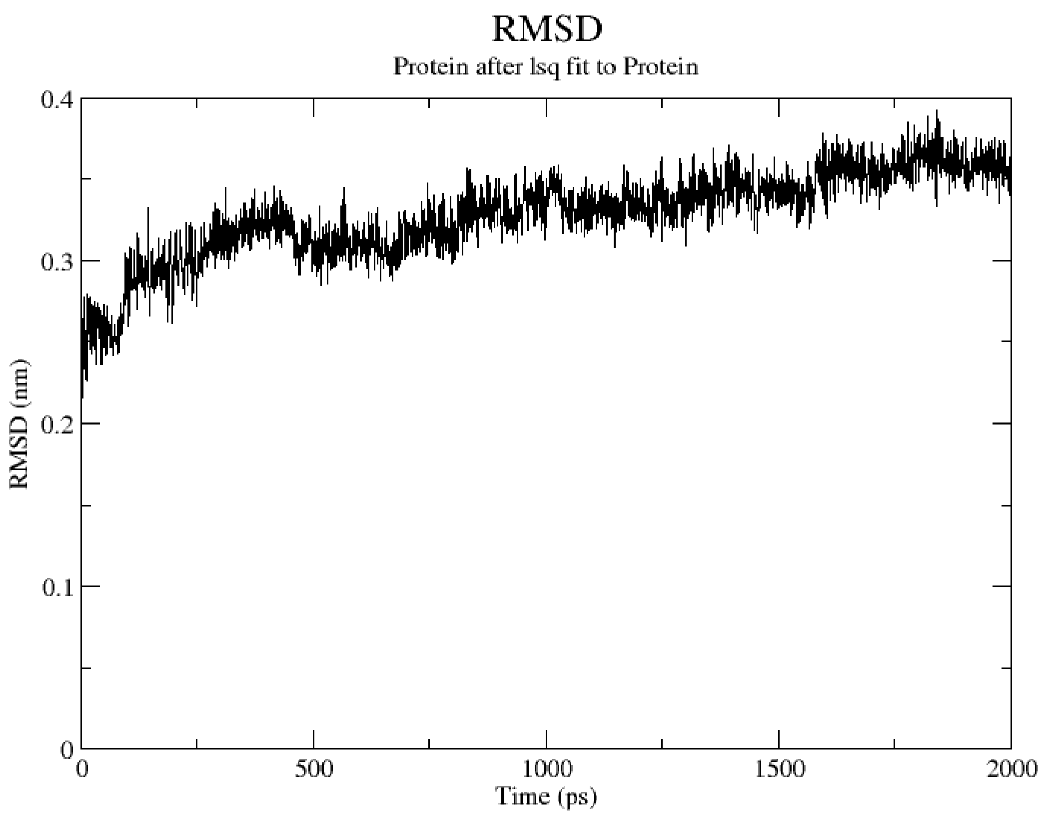
```
(base) eduardoruiz@192 md_water % grep "Performance" md.log
Performance:       43.079        0.557
(base) eduardoruiz@192 md_water %
```

By removing 2,214 water molecules (6,642 atoms), the total number of pair interactions the processor must compute at every integration step is drastically reduced. This allows the system to advance through the 2 nanosecond trajectory more than twice as fast as the solvated system.

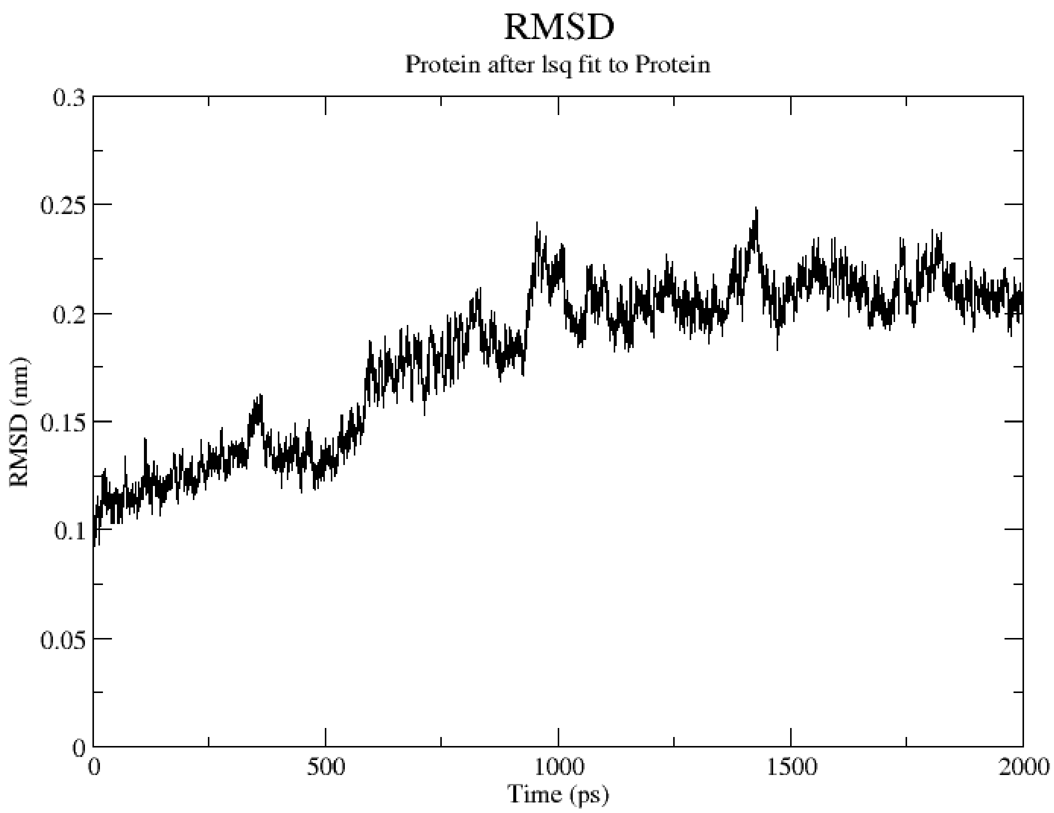
**2) How stable is the protein when it moves in vacuum compared to when it is in solution, in terms of RMSD and radius of gyration?**

The protein is markedly more stable in solution than in vacuum. In water, both RMSD and radius of gyration reach stable plateaus, indicating preservation of the native folded structure. In contrast, in vacuum the RMSD continuously increases and the radius of gyration shows larger fluctuations and drift, revealing progressive structural destabilization and expansion due to the absence of solvent stabilization

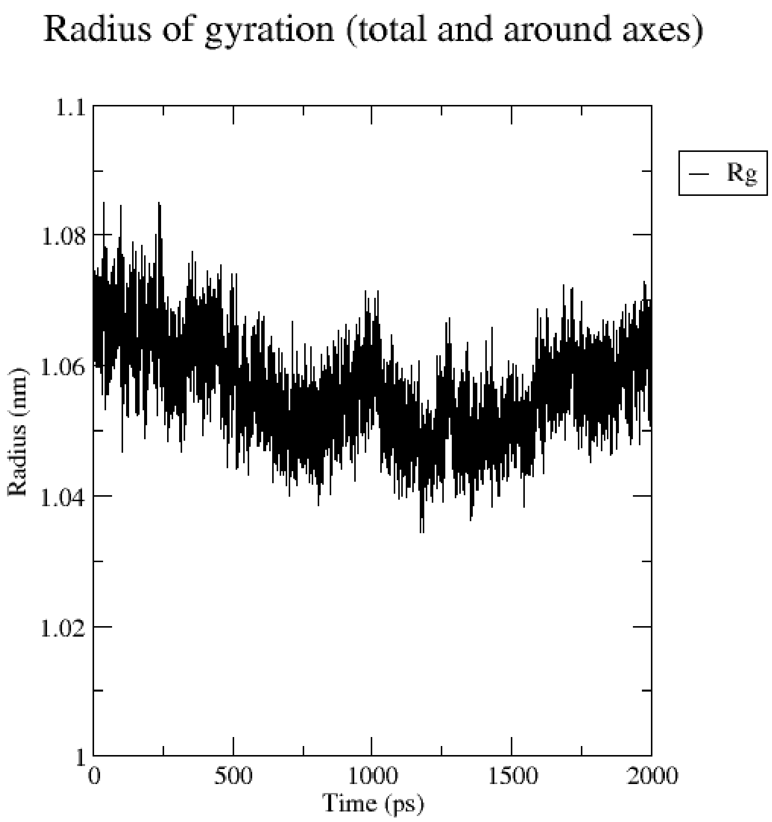
**vacuum:**



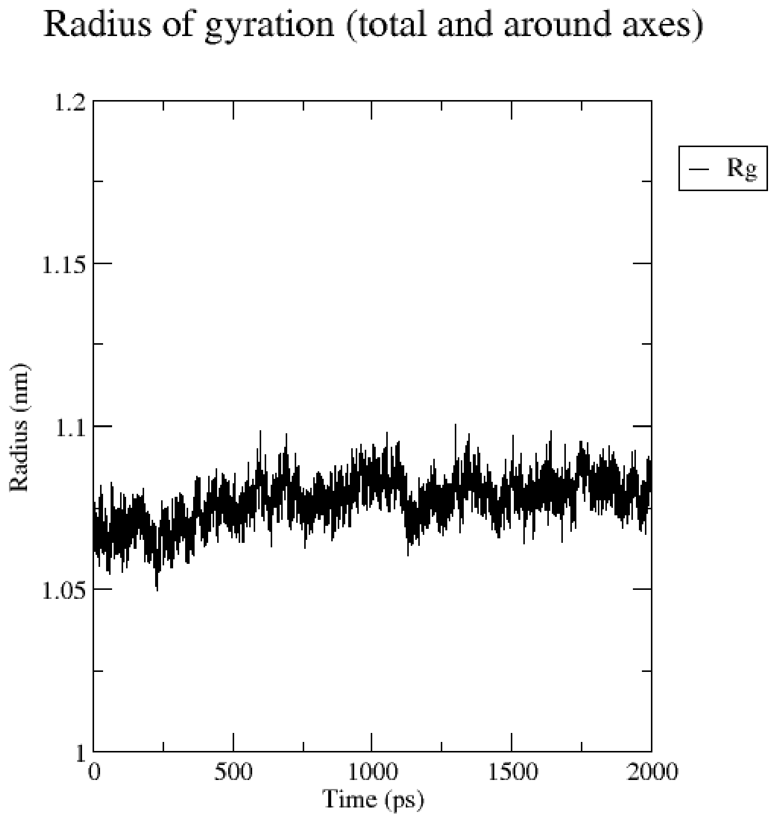
**water:**


RMSD
Protein after lsq fit to Protein
0.3
0.25
0.2
0.15
0.1
0.05
0
RMSD (nm)
0
500
1000
1500
2000
Time (ps)

**vacuum:**


Radius of gyration (total and around axes)
1.1
1.08
1.06
1.04
1.02
1
Radius (nm)
0
500
1000
1500
2000
Time (ps)
Rg

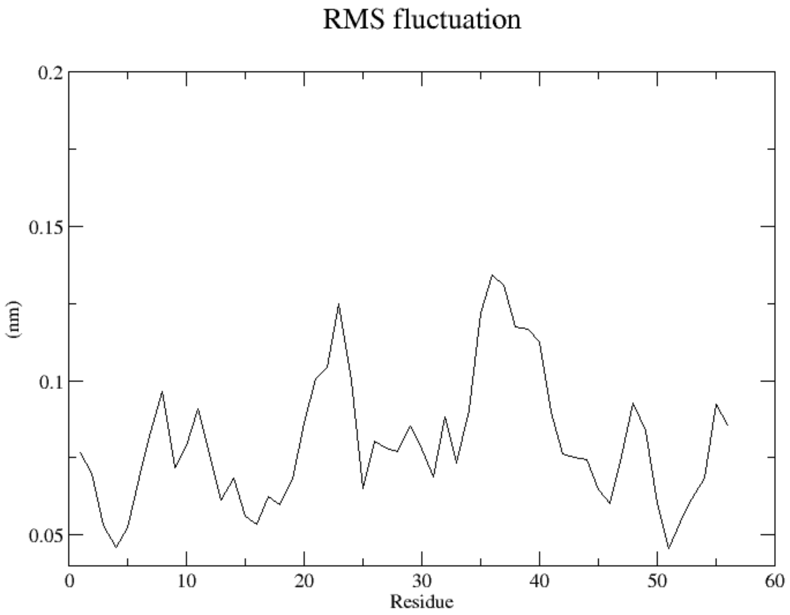
**water:**



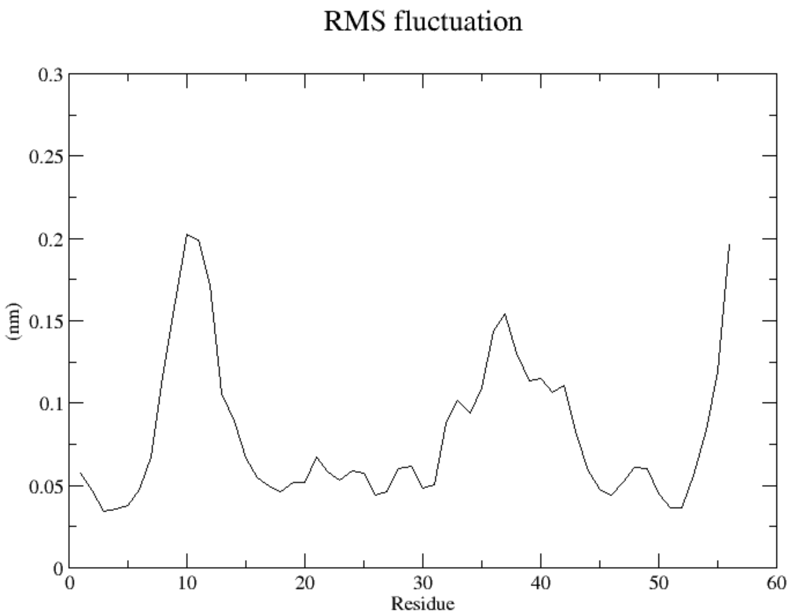

**3) Determine if the lack of solvent caused a change in the flexibility of certain parts of the protein, monitoring the RMSF, and if that is the case, explain why.**

The RMSF (Root Mean Square Fluctuation) reveals that the lack of solvent significantly alters the protein's local flexibility. In a vacuum, the protein tends to collapse upon itself to maximize internal interactions, resulting in more uniform and constrained fluctuations as the structure "locks" into a compact state. In contrast, the water environment facilitates much higher flexibility in specific regions (notably around residue 10 and the C-terminus), as the solvent acts as a molecular lubricant. This occurs because water molecules form competitive hydrogen bonds with the protein surface, breaking rigid internal constraints and allowing loops and surface residues to sample a wider range of motion.

**vacuum:**



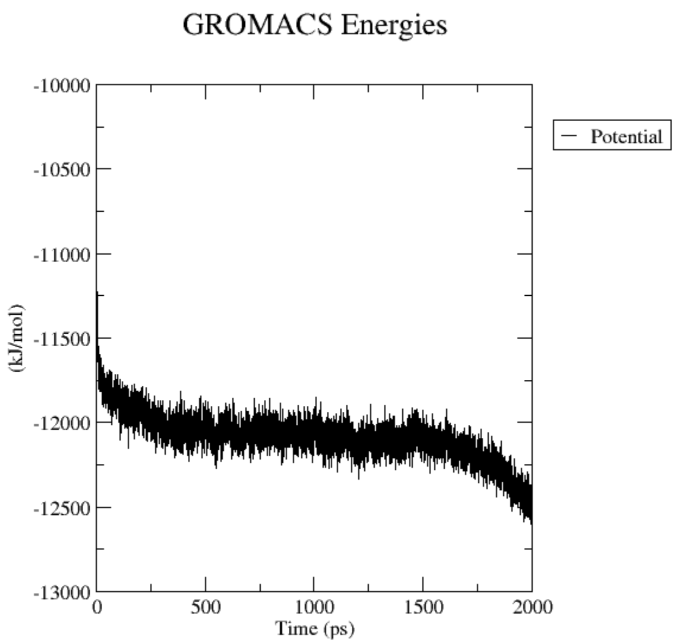

**water:**



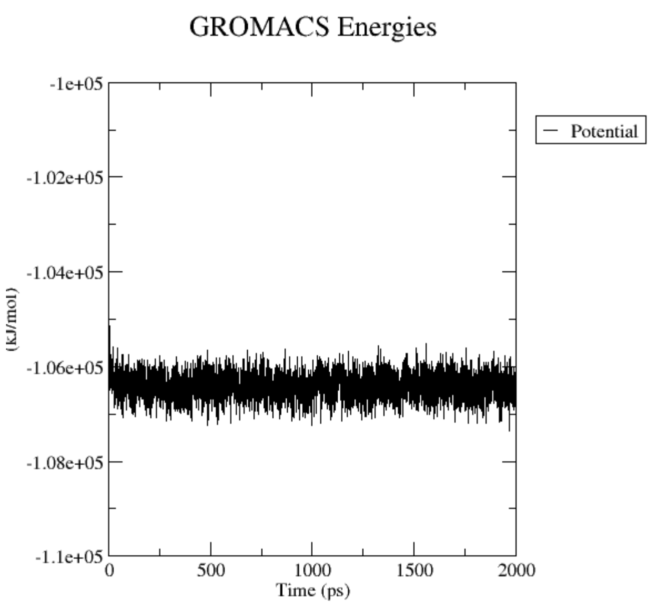

**4) How does the potential energy compare in the solvated and in the vacuum cases? provide an explanation of the observed behavior.**

The potential energy in the water case is roughly ten times more negative than in the vacuum case. This is because the total potential energy of the solvated system includes not only the protein's internal interactions but also the thousands of new interactions between the protein and water molecules, as well as water-water interactions (hydrogen bonding). In vacuum, the protein only has its own atoms to interact with, leading to a much higher (less negative) energy state.

**vacuum.**

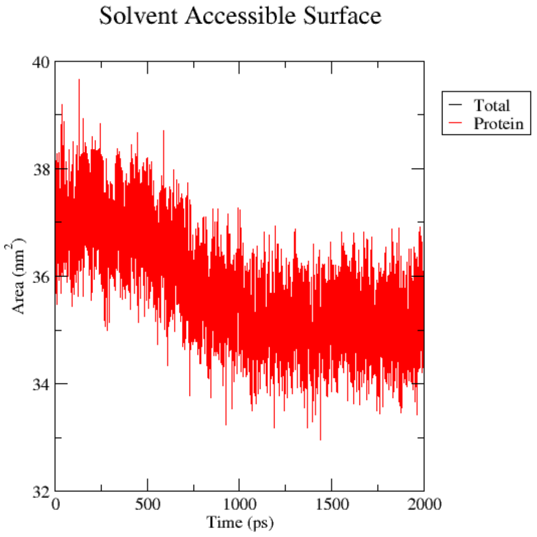



**water:**



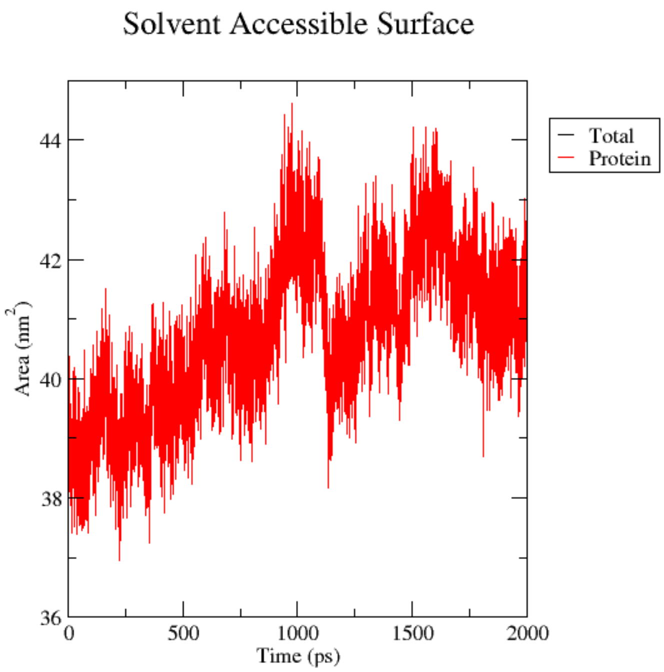
**5) What can you say about the hydrophobic area exposed to the solvent and to the vacuum?**

**vacuum:**




In the vacuum simulation, the total surface area decreases from approximately 37 $nm^2$ to 35$nm^2$ over the 2000 ps trajectory. This confirms the hydrophobic collapse. Without water molecules to interact with, the protein's hydrophobic residues hide by packing as tightly as possible toward the center of the structure to maximize internal van der Waals interactions. This results in a more compact, less flexible, and non native conformation.

**water:**




In the aqueous environment, the surface area actually increases, stabilizing at a much higher value of roughly 41-42 $nm^2$. This suggests that the water environment encourages the protein to maintain its native, expanded fold.

The lack of solvent in the vacuum case forces the protein to sacrifice its functional surface area to achieve internal stability. In contrast, the water environment provides the necessary hydrogen bonding partners to allow the protein to expose a larger surface area without an energetic penalty, maintaining the structural integrity required for biological activity.

**6) Why do you think that it is important to include explicit solvent in the simulation of a protein?**

Including explicit solvent is crucial because proteins naturally exist and function in water, not in vacuum. Water is not just a background medium, it actively stabilizes the protein structure through hydrogen bonding, electrostatic screening, and hydrophobic effects. In water, polar and charged residues interact with solvent molecules, which helps the protein adopt and maintain a realistic folded conformation. Water also provides friction and damping, allowing the protein to relax smoothly and explore conformations that resemble real biological conditions.
Without solvent, electrostatic interactions become unrealistically strong, hydrophobic residues are no longer driven to stay buried, and the protein often becomes artificially compact or behaves in a non physical way. As a result, simulations in

vacuum may look numerically stable but do not represent how proteins behave in living systems.

annexes: